Energy management

Our ambition is to increase efficiency despite increasing production. By continuously monitoring and analysing energy consumption, we can identify energy hotspots. Our sites in Ellwangen, Nördlingen and Brasov are certified in accordance with ISO 50001 standards for energy management systems for this purpose. Energy Management Systems. In addition, an improved energy management system to implement the most effective measures to achieve our efficiency and reduction targets will be introduced in 2022.

The energy used to produce batteries is associated with high environmental impacts derived from electricity production. We believe that sourcing low-carbon energy is essential for a successful energy transition. Thus, we have been using green electricity at all of our production sites in Europe since 2021. We are also increasing our own production of renewable energy by installing two further photovoltaic systems, which will become operational in 2022. With our existing PV systems, we are able to produce up to 1,000 MWh of electricity per year.

Goal 5: Striving for energy-efficiency

i

Through the commissioning of further photovoltaic systems, we can increase our PV capacity **7-fold**.

Expanding production capacities, constructing new plants and the increased production of rechargeable batteries – a more energy intensive production process – have led to an overall higher energy consumption in 2021.

VARTA AG's energy consumption in MWh*

LOCATION YEAR	2019	2020	2021
ELLWANGEN	27,321	30,977	33,347
NÖRDLINGEN	5,303	12,962	21,748
DISCHINGEN/ NEUNHEIM	18,810	18,017	19,035
BRASOV	1,707	1,985	2,883
BATAM	2,357	2,241	2,074
TOTAL	**55,498**	**66,182**	**79,087**
ENERGY / EMPLOYEE	**20**	**14**	**17**

* Excluding sales offices and non-production sites contributing less than approximately 1% as well sites without operational control.

Waste management

We continuously optimise our processes to minimise waste and reduce, reuse or regrind materials within the production processes. The remaining waste from production is separated and processed by qualified waste management companies. In 2021 we generated 4,648 tons of waste in total; of this figure, 2,755 tons were recycled.

VARTA AG's waste and proportion of recycled waste in tons*

LOCATION (t)	WASTE	PROPORTION OF RECYCLED WASTE
ELLWANGEN	1,711	1,143
NÖRDLINGEN	1,038	292
DISCHINGEN / NEUNHEIM	1,439	1,046
BRASOV	374	251
BATAM	86	23
TOTAL	**4,648**	**2,755**

VARTA AG's water consumption in m³*

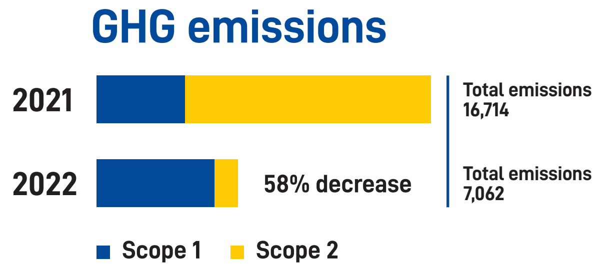
LOCATION YEAR	2019	2020	2021
ELLWANGEN	11,899	17,359	17,215
NÖRDLINGEN	2,516	2,989	3,252
DISCHINGEN / NEUNHEIM	7,993	9,203	8,468
BRASOV	2,275	2,679	3,045
BATAM	13,649	13,695	14,204
TOTAL	**38,332**	**45,925**	**46,184**
CONSUMPTION / EMPLOYEE	**14**	**10**	**10**

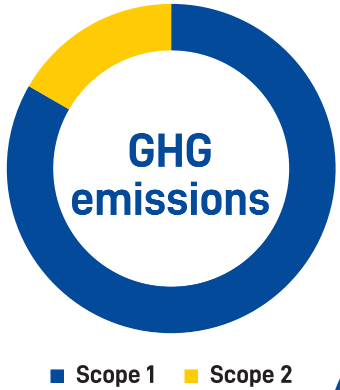
Water management

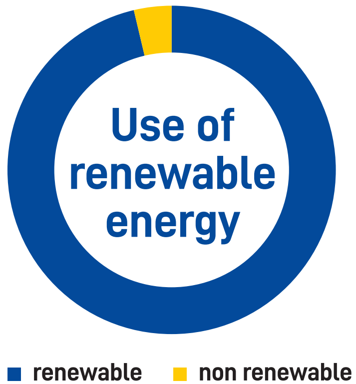
Water consumption is monitored regularly. If anomalies occur, causes are identified and control measures initiated. We mainly consume water through the social and sanitary facilities at our sites. In production, water is required during the cleaning processes for the end products and for our galvanic equipment. For these, water treatment plants are installed downstream to remove impurities and ensure high water quality.

Greenhouse gas emissions

To effectively combat climate change, it is essential to know our own contribution. Therefore, we account for our greenhouse gas emissions according to the GHG Protocol. Only by accurately determining our own emissions we can identify hotspots and take effective measures to reduce them. Our measures have resulted in a 58% decrease in our GHG emissions (Scope 1 and Scope 2) compared to the previous year. Please refer to page 39 for more detail.







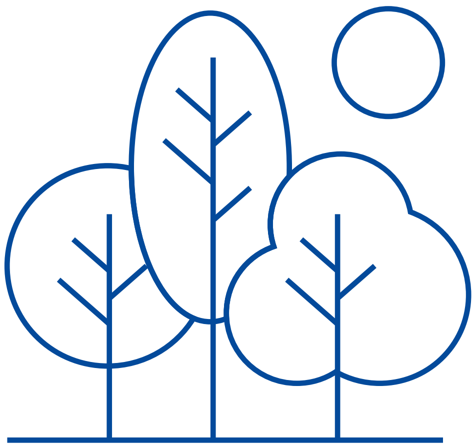
* Excluding sales offices and non-production sites contributing less than approximately 1% as well sites without operational control.

Shaping sustainable solutions for the planet

Taking responsibility for our direct environment

In most recent years, our efforts were focused on a forest area adjacent to one of our sites in Germany. This forest is a protected natural monument and biotope. As an insect population posing a health risk to humans and the forest itself was identified, measures were coordinated and approved by local and governmental authorities to countermeasure the negative impact. In 2021 the measures proved to be effective and mitigated the threat.

We bear responsibility for the regions and natural habitats surrounding our sites to minimise negative influences and contribute towards positive change.



… will save **2kg plastic**/1,000 batteries

… is 100% recyclable

Cardboard packaging

FSC-certified cardboard
Made of up to 95% recycled content
100% recyclable

… will save **4.4kg CO_2** emissions/1,000 batteries

… is recyclable

Dual-material packaging

FSC-certified cardboard
Made of up to 90% recycled content
Recyclable

Eco-friendly packaging

This year, we introduced our new VARTA logo together with our raised brand profile. In this process, all our VARTA Consumer products have been comprehensively updated. Now each individual category features significant product improvements, a new haptic packaging experience and thoughtfully conceived sustainable packaging solutions.

With our new packaging, we are continuing our efforts to conserve resources. Thus, we have eliminated plastic as a packaging material for many products and ensured the ink used for our packaging is environmentally compatible. Our cardboard packaging also features the innovative, patent-pending opening/closing mechanism that allows cells to be removed one at a time. Moreover, we have increased the recycled content from 90% to 95% in the cardboard packaging compared to the dual-material packaging.

Currently, 21% of the products in our VARTA Consumer portfolio are sold in our new environmentally-friendly packaging. As a result, we were able to reduce plastic consumption at VARTA Consumer by 15% in 2021 and are aiming for a 35% reduction by 2023. To ensure the necessary capacities are available, we have invested €6 m whith a further €14 m earmarked for this project.

In the future, we will be expanding this to cover all VARTA operations. Moreover, we continue our efforts to develop new intelligent and childproof packaging solutions.

Annex

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Methodology

Total energy management

Total energy consumption covers electricity consumption as well as gas consumption for the generation of (process) heat and fuels. Corresponding data were determined by means of invoices or meter readings. Presented data covers all our production sites. Sales offices and non-production sites as well sites without operational control are excluded and would contribute less than approximately 1%[1].

Water management

Water consumption was recorded by means of water meters or, if this was not possible, by invoicing. The figures in table 3 exclude sales offices and non-production sites that are expected to contribute to less than approximately 2%[2] as well sites without operational control.

Waste

Generated waste includes both waste from production and administrative facilities. Only waste that we are certain can be recycled by our waste management service providers is included in recycled waste. The respective quantities were determined on the basis of invoices. Waste from sales offices, non-production sites and sites without operational control would contribute to less than approximately 6%[3] and are not included in the data presented.

GHG emissions

This year, we focused on determining Scope 1 and 2 emissions. Over the course of the next year, we will advance to include Scope 3.

VARTA AG's GHG emissions in CO_2 eq t

SCOPE YEAR	2019	2020	2021
Scope 1[4]	3,598	4,417	5,894[5]
Scope 2[6]	10,347	12,296	1,168
TOTAL EMISSIONS	**13,945**	**16,714**	**7,062**
USE OF BIOMASS	**0**	**54**	**243**

This GHG inventory was compiled in accordance with the WRI/WBCSD Greenhouse Gas (GHG) Protocol.

In accordance with the GHG Protocol Corporate Accounting Standard, we report Scope 1 and Scope 2 emissions separately. For Scope 2, we consider the marked-based approach as well as the location-based approach.

We report on all emissions from our production facilities worldwide. The organisational boundary applied to consolidate VARTA's emissions was the financial control approach.

[1] The Energy Institute for Business (Energieinstitut für Wirtschaft), 2012, 'Energiekennzahlen in Dienstleistungsgebäuden', Vienna, https://www.energieinstitut.net/de/system/files/0903_final_dienstleistungsgebaude_20120530.pdf [accessed 1 March 2022].
[2] South Staffordshire Water Plc, n.d., 'Water use in your business', UK, https://www.south-staffs-water.co.uk/media/1509/waterusebusiness.pdf, [accessed 1 March 2022].
[3] Estimation based on ratio of employees in included and excluded sites.
[4] Emissions of N_2O, CH_4 and HFC have been translated into CO_2 emissions using the Global Warming Potential, or GWP, factor. GWP factors are based on the Intergovernmental Panel on Climate Change (IPCC) Fifth IPCC Assessment Report (AR5) using 100-year values. The latest (AR5) values are used for the current inventory.
[5] Due to construction work at one site, emergency generators had to be used, which were operated with diesel fuel.
[6] Scope 2 emissions calculated by using the market-based approach. The location-based approach would result in the following figures: 17,038 CO_2-eq t (2019), 20,885 CO_2-eq t (2020), and 23,503 CO_2-eq t (2021). In accordance with the WRI/WBCSD Greenhouse Gas (GHG) Protocol.

Some minor non-production facilities have been excluded from the inventory boundary over the reporting period FY19-21, due to lack of data.

For FY19, there is no emission data for the refrigerants from the production sites Brasov and Batam.

Assumptions were made regarding fuel consumption for the vehicle fleet at the locations. The total fuel costs for company and pool vehicles in a fiscal year were offset against the average fuel prices in the same year.

Scope 1 GHG emissions –
Activity data and emissions include on-site stationary burning of fossil fuels (e.g., boilers) or process-related emissions (e.g., backup generators), company-owned or leased vehicles. Fugitive emissions associated with the use of HVAC equipment are included here.

Scope 2 GHG emissions (location based) –
According to the Scope 2 guidance of the GHG Protocol, VARTA uses the national or regional emission factors for indirect (Scope 2) emissions, which are defined by the following methods in each relative region in which VARTA operates:

- German Federal Environment Agency (UBA) from Fuel Combustion
- For Romanian sites: Austrian Energy Agency[7]
- For Indonesian sites: Climate Transparency[8]

Scope 2 GHG emissions (marked based) –
Based on the latest available emission factors published by the electricity supplier(s) that relate specifically to the CO_2 intensity of the electricity purchased.

Market-based emission factors including supporting evidence (e.g., energy attribute certificates, supplier invoices, etc.) for the reporting years 2019-21 were collected by VARTA's Energy Management Officer and evaluated based on the criteria as described in the GHG Protocol standard requirements for Scope 2 reporting.

Due to the nature of VARTA's operations, all relevant gases that occur in significant quantities are tracked. Global Warming Potentials (GWPs) references are from the Intergovernmental Panel on Climate Change (IPCC) Fifth IPCC Assessment Report (AR5) using 100-year values. The latest (AR5) values are used for the current inventory.

FY19-21 is the baseline period for future reporting. Data for FY19-21 across the Group is the best available data from VARTA AG.

Calculation of EU-taxonomy indicators

To identify our taxonomy-eligible and non-eligible business activities, a project team consisting of experts in the field of legal, finance and sustainability analysed all business activities and assigned them to the relevant category. Afterwards, a further analysis of the taxonomy eligibility of the individual activities was carried out with experts from the departments in question. Finally, we structured our activities in sales revenue, capital expenditure (CapEx) and operating expenditure (OpEx).

Taxonomy-eligible economic activities at VARTA include battery solutions from our business segments Power Pack Solutions, Energy Storage Systems and Large Cells. For the calculation, revenue is defined as net sales in accordance with IFRS, as reported in the consolidated income statement. Further information can be found in the annual report on page 44 . CapEx is calculated on a gross basis, without taking into account revaluations or scheduled or unscheduled depreciation. It includes investments in non-current intangible or tangible assets, including assets acquired through asset or share deals, as shown in the consolidated statement of financial position (see page 50 and page 76 of the Annual Report). OpEx, on the other hand, includes non-capitalisable expenses recognised in the consolidated statement of income, such as research and development, building refurbishment measures, short-term leasing, maintenance and repair as well as all other direct expenses arising from the upkeep of property, plant and equipment to ensure that the taxonomy-eligible assets are ready for operation.

For the 2022 financial year, VARTA will additionally have to report on the taxonomy conformity of taxonomy-eligible products. As things stand, all products classified as taxonomy-eligible under criteria set 3.4 will also be taxonomy-compliant. Further checks are still required for products that have been classified as taxonomy-eligible under criteria set 3.6.

[7] Energyagency.at, Vienna, Austrian Energy Agency, n.d., https://www.energyagency.at/ [accessed 1 March 2022].
[8] Climate Transparency, 2018, 'Brown to Green: The G20 Transition to a Low-carbon Economy', Climate Transparency, Berlin, Germany, p.7 [accessed 15 March 2022].

GRI content index

Statement of use	VARTA AG has reported the information cited in this GRI content index for the period 01 January to 31 December 2021 with reference to the GRI Standards.
GRI 1 used	GRI 1: Foundation 2021

GRI STANDARD & DISCLOSURE	LOCATION & COMMENTS
GRI 2: General Disclosures 2021	
2–1 (a, b) Organizational details	VARTA AG is a publicly traded company and has been listed on the Frankfurt Stock Exchange in the Prime Standard segment since October 2017, in addition to being included on the select indices of the MDAX and TecDAX since 23 December 2019.
2–2 (a–c) Entities included in the organization's sustainability reporting	All entities of VARTA AG as per 31/12/2021 were included. The control approach is used for sustainability reporting. Thus, subsidiaries or companies where VARTA holds a majority stake have been included unless otherwise stated. Please refer to page 151 et seq. in the Annual Report.
2–3 (a–d) Reporting period, frequency and contact point	The report covers topics from the financial year 2021, which extends from 01 January to 31 December 2021 and is published annually. The Sustainability Report 2021 was published on 05/06/2022. For questions about the report or more information, please contact us at sustainability@varta-ag.com.
2–6 (a, b) Activities, value chain and other business relationships	Company overview Business overview Sustainable supply chain
2–7 (a) Employees	Employees
2–9 (a, b) Governance structure and composition	Governance structure Annual Report p. 7–14 Declaration of Conformity
2–11 (a, b) Chair of the highest governance body	Annual Report p. 7–14 Declaration of Conformity
2–13 (a, b) Delegation of responsibility for managing impacts	Governance structure
2–14 (a, b) Role of the highest governance body in sustainability reporting	The sustainability report is reviewed and approved by the Executive Board.
2–15 (a, b) Conflicts of interest	Annual Report p. 13 Declaration of Conformity
2–19 (a, b) Remuneration policies	Annual Report p. 147, 152 Remuneration Report
2–20 (a, b) Process to determine remuneration	Remuneration Report
2–22 (a) Statement on sustainable development strategy	A promise we keep
2–23 (a–f) Policy commitments	Policy for responsible sourcing Due Diligence Report 2021
2–24 (a) Embedding policy commitments	Policy for responsible sourcing Due Diligence Report 2021
2–26 (a) Mechanisms for seeking advice and raising concerns	Business ethics